

Samiur R. Mir

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EDUCATION

Oklahoma State University

Ph.D. in Physics

Advisor: Prof. Vedran Brdar

Oklahoma, United States

Aug. 2021 – Present

University of Alberta

M.Sc. in Physics

Alberta, Canada

Sept. 2019 – Apr. 2021

University of Dhaka

B.Sc. in Physics

Dhaka, Bangladesh

Jan. 2015 – Aug. 2019

PUBLICATIONS & PREPRINTS

6. Vedran Brdar, **Samiur R. Mir**, Xun-Jie Xu, “*Solar Constraints on Heavy Neutral Leptons with ν_τ Mixing*”, [arXiv:2608.19319 \[hep-ph\]](#) (under PRL review)
5. Vedran Brdar, **Samiur R. Mir**, “*Ultra-High-Energy Tau Neutrinos as Probes of Lorentz Invariance*”, [arXiv:2604.19880 \[hep-ph\]](#) (under JCAP review)
4. Vedran Brdar, Dibya S. Chattopadhyay, **Samiur R. Mir**, Tousif Raza, and Marc S. Romanowski, “*Prospects for Neutrino Observation and Mass Measurement from Binary Neutron Star Mergers*”, *Phys. Rev. D* **113**, 123017 (2026), [arXiv:2511.16658 \[hep-ph\]](#)
3. **Samiur R. Mir**, Carlos A. Argüelles, K.S. Babu, Vedran Brdar, “*Double Bangs in IceCube as a Window to the Neutrino Mass Origin*”, *Phys. Lett. B* **875** (2026) 140322, [arXiv:2511.07541 \[hep-ph\]](#)
2. Vedran Brdar, Ying-Ying Li, **Samiur R. Mir**, Yi-Lin Wang, “*Collider Prospects for the Neutrino Magnetic Moment Portal*”, *JHEP* **10** (2025) 230, [arXiv:2502.07024 \[hep-ph\]](#)
1. M. Jamil Aslam, Wen Chen, Andrzej Czarnecki, **Samiur R. Mir**, Muhammad Mubasher, “*Rare Decays of the Positronium Ion and Molecule, $\text{Ps}^- \rightarrow e^- \gamma$ and $\text{Ps}_2 \rightarrow e^+ e^- \gamma, \gamma \gamma, e^+ e^-$* ”, *Phys. Rev. A* **104**, 052803 (2021), [arXiv:2108.06785 \[hep-ph\]](#)

INVITED TALKS

RG-Running Neutrino Mixing at IceCube

2026

Latin American Webinars in Physics (Webinar 195, [available on YouTube](#))

Probing Neutrinophilic BSM Scenarios with High-Energy Neutrinos

2026

HEP seminar, PITT PACC, University of Pittsburgh

Unveiling the Neutrino Magnetic Moment: Constraints from Colliders

2025

LPPC seminar, Harvard University

CONFERENCE PRESENTATIONS

Testing Lorentz Invariance with Ultra-High-Energy Tau Neutrinos	2026
SUSY Conference, University of Oklahoma	
Testing RG Evolution of Neutrino Mixing with IceCube Double Bang Events	2026
Pheno Symposium, University of Pittsburgh	
Probing Neutrino Mass Models with Tau Neutrino Appearance at IceCube	2025
Particle Physics on the Plains, University of Kansas	
Probing Neutrino Mass Models with Tau Neutrino Appearance at IceCube	2025
APS Meeting Texas, Arkansas, Oklahoma Section, Texas Christian University	
Unveiling the Neutrino Magnetic Moment: Constraints from Colliders	2025
Pheno Symposium, University of Pittsburgh	
Unveiling the Neutrino Magnetic Moment: Constraints from Colliders	2025
LPPC seminar at Harvard University	
Neutrino Magnetic Moment at Colliders	2024
Particle Physics on the Plains, University of Kansas	
Collider Probes of Neutrino Magnetic Moment	2024
DPF - Pheno Symposium, University of Pittsburgh	
Single Photon Decay of the Positronium Ion	2020
57th Winter Nuclear and Particle Physics Conference, Banff, Alberta	
How to Find Decay Distribution from Symmetry	2019
10th Annual Symposium for Graduate Physics Research, University of Alberta	

SUMMER SCHOOLS

Cosmic Neutrino Practicum <i>Harvard University</i>	June 2026
International Neutrino Summer School <i>Fermilab</i>	Aug. 2025
NVIDIA Fundamentals of Deep Learning workshop <i>Oklahoma State University</i>	June 2023

ACHIEVEMENTS

Student Travel Awards <i>granted by APS Fall Meeting</i>	2025
Student Travel Awards <i>granted by APS and Pheno organizers</i>	2024
Recipient of Honours Scholarship (top 5% CGPA in Bachelors) <i>University of Dhaka</i>	2019
National Olympiad Champion <i>Bangladesh National Math Olympiad</i>	2013

MENTORING EXPERIENCE

Safius Sakib Shuddho , University of Dhaka	Aug. 2026–Oct. 2026
ICTP PWF Physics For Bangladesh Internship	
Marc Romanowski , The College of New Jersey	June 2025–Aug. 2025
NSF REU Visiting Undergraduate Researcher	

LEADERSHIP ACTIVITIES

Coordinator and Contributor, HEP-pheno Student Seminar	2023 – 2025
<i>Dept. of Physics, Oklahoma State University</i>	
Vice-President, Physics Graduate Student Association	2023 – 2025
<i>Dept. of Physics, Oklahoma State University</i>	
Outreach Volunteer, Physics Open House	2023 – Present
<i>Dept. of Physics, Oklahoma State University</i>	
Academic Team Member <i>Bangladesh Mathematical Olympiad</i>	2015 – 2019
Academic Instructor <i>Community of Physics, Bangladesh</i>	2015 – 2019

TEACHING EXPERIENCE

Oklahoma State University	
Teaching Assistant: Electricity & Magnetism (Labs & Discussions)	Aug. 2021 –Present
University of Alberta	
Teaching Assistant: Particles and Waves, Electricity and Magnetism (Labs)	Aug. 2019–April 2021